Rafael Petrosian

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WORK EXPERIENCE

EDN Aviation **Jun. 2025 – Aug. 2025**

Assistant Engineer *Van Nuys, CA*

- Led a 1-month electrical design initiative under the guidance of Senior Engineer, successfully transcribing an electrical plan into CircuitJS and developing a test box; overcame schematic modeling challenges with team collaboration.
- Developed precise CAD models using OnShape by interpreting mechanical plans and performing accurate measurements, directly reducing CNC manufacturing errors.
- Streamlined procurement by preparing extensive and detailed Bills of Materials (BOMs), supporting regular client quotation processes.
- Automated the packaging and labeling workflow, saving 15% of man-hours by scripting unique QR code generation and eliminating manual processes.

πRo-Bot: Autonomous Fire Suppression Robot **Jan. 2025 – May. 2025**

Lead Programmer *Berkeley, CA*

- Developed an automated fire suppression system capable of detecting, extinguishing, and preventing fires.
- Spearheaded the electrical system design encompassing circuitry, power distribution, and microcontroller integration.
- Utilized Arduino IDE and C++ to write a closed-loop system for an autonomous fire suppression protocol.
- Collaborated with team by designing and iterating through digital twins to find the optimal electromechanical design.

Cal Aero SAE **Sep. 2023 – Apr. 2024**

Wing Design Engineer *Berkeley, CA*

- Led a team to design, build, test, and fly an RC plane, achieving a top 15 ranking in the Aero SAE competition over 1 year, gaining hands-on experience in prototyping and data-driven design decisions.
- Employed MATLAB to model and verify spar sizing through data-driven methods, achieving stiffness/strength targets and reducing material usage by 10% while ensuring manufacturability.
- Leveraged Python scripts to verify optimal airfoil design, leading to a CAD model development in SolidWorks for the final wing assembly, enhancing aerodynamic efficiency.
- Conducted fluid-dynamics analysis through XFLR5 to optimize dihedral and wing position, achieving a superior balance between stability and maneuverability.

Thermal Paste Performance Experiment **Aug. 2024 – Dec. 2024**

Analyst *Berkeley, CA*

- Designed and executed a 4-month controlled heat-transfer experiment to evaluate CPU thermal pastes, utilizing custom-machined aluminum blocks and advanced data acquisition with K-type thermocouples and ESP32 DAQ
- Processed and smoothed multiple data sets from thermocouples calibrated in MATLAB, ensuring consistent and accurate ranking of thermal pastes using ΔT as the performance metric under repeatable conditions
- Created a custom experimentation setup from 6061 aluminum using milling techniques, assembled with insulation to ensure precise thermal testing conditions

EDUCATION

University of California, Berkeley **Dec. 2025**

Bachelor of Science, Mechanical Engineering *Berkeley, CA*

SKILLS, TECHNOLOGIES, LANGUAGES & INTERESTS

- **Skills:** Feedback Control Systems; Additive Manufacturing; Mechatronics; Programming; Electrical Design; Aerodynamic Modeling; CAD Modeling and Design; Thermal Systems; Data Administration and Validation; Autonomous Systems; Circuit Design and Validation; Field Instrumentation; Motor Control
- **Technologies:** MATLAB; ESP Microcontroller; Firmware; Python; SOLIDWORKS; C++; Geometric Dimensioning & Tolerancing; OnShape; Fusion 360; Mill and Lathe
- **Languages:** English; Armenian
- **Interests:** Music Production; Traveling; Cars; Chess; Movies